SAT Math

Equivalent Expressions 3

Question #	ID		Answer
3.1	371cbf6b	$(ax+3)(5x^2-bx+4)=20x^3-9x^2-2x+12$ The equation above is true for all x, where a and b are constants. What is the value of ab ? A. 18 B. 20 C. 24 D. 40	C
3.2	40c09d66	If $\dfrac{\sqrt{x^5}}{\sqrt[3]{x^4}}=x^{\frac{a}{b}}$ for all positive values of x, what is the value of $\dfrac{a}{b}$?	7/6
3.3	34847f8a	$\dfrac{2}{x-2}+\dfrac{3}{x+5}=\dfrac{rx+t}{(x-2)(x+5)}$ The equation above is true for all $x>2$, where r and t are positive constants. What is the value of rt ? A. -20 B. 15 C. 20 D. 60	C
3.4	137cc6fd	$\sqrt[5]{70n}\left(\sqrt[6]{70n}\right)^2$ For what value of x is the given expression equivalent to $(70n)^{30x}$, where $n>1$?	4/225

Question #	ID		Answer
3.5	ea6d05bb	The expression $(3x-23)(19x+6)$ is equivalent to the expression ax^2+bx+c, where a, b, and c are constants. What is the value of b?	-419
3.6	d8789a4c	$\frac{x^2-c}{x-b}$ In the expression above, b and c are positive integers. If the expression is equivalent to $x+b$ and $x \neq b$, which of the following could be the value of c ? A. 4 B. 6 C. 8 D. 10	A
3.7	5355c0ef	$$0.36x^2+0.63x+1.17$$ The given expression can be rewritten as $a\left(4x^2+7x+13\right)$, where a is a constant. What is the value of a?	0.09
3.8	c81b6c57	In the expression $3(2x^2+px+8)-16x(p+4)$, p is a constant. This expression is equivalent to the expression $6x^2-155x+24$. What is the value of p ? A. -3 B. 7 C. 13 D. 155	B

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3.9	2c88af4d	The expression $\dfrac{x^{-2}y^{\frac{1}{2}}}{x^{\frac{1}{3}}y^{-1}}$, where $x > 1$ and $y > 1$, is equivalent to which of the following? A. $\dfrac{\sqrt{y}}{\sqrt[3]{x^2}}$ B. $\dfrac{y\sqrt{y}}{\sqrt[3]{x^2}}$ C. $\dfrac{y\sqrt{y}}{x\sqrt{x}}$ D. $\dfrac{y\sqrt{y}}{x^2\sqrt[3]{x}}$	D
3.10	22fd3e1f	$f(x) = x^3 - 9x$ $g(x) = x^2 - 2x - 3$ Which of the following expressions is equivalent to $\dfrac{f(x)}{g(x)}$, for $x > 3$? A. $\dfrac{1}{x+1}$ B. $\dfrac{x+3}{x+1}$ C. $\dfrac{x(x-3)}{x+1}$ D. $\dfrac{x(x+3)}{x+1}$	D
3.11	a0b4103e	The expression $\dfrac{1}{3}x^2 - 2$ can be rewritten as $\dfrac{1}{3}(x-k)(x+k)$, where k is a positive constant. What is the value of k? A. 2 B. 6 C. $\sqrt{2}$ D. $\sqrt{6}$	D

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3.12	ad038c19	Which of the following is equivalent to $\left(a+\frac{b}{2}\right)^2$?	D

A. $a^2+\frac{b^2}{2}$

B. $a^2+\frac{b^2}{4}$

C. $a^2+\frac{ab}{2}+\frac{b^2}{2}$

D. $a^2+ab+\frac{b^2}{4}$

3.13	12e7faf8	The equation $\frac{x^2+6x-7}{x+7}=ax+d$ is true for all $x\neq-7$, where a and d are integers. What is the value of $a+d$?	C

A. -6

B. -1

C. 0

D. 1

3.14	89fc23af	Which of the following expressions is equivalent to $\frac{x^2-2x-5}{x-3}$?	D

A. $x-5-\frac{20}{x-3}$

B. $x-5-\frac{10}{x-3}$

C. $x+1-\frac{8}{x-3}$

D. $x+1-\frac{2}{x-3}$

Question #	ID		Answer
3.15	911c415b	$(7532+100y^2)+10(10y^2-110)$ The expression above can be written in the form ay^2+b , where *a* and *b* are constants. What is the value of $a+b$?	6632
3.16	f89e1d6f	If $a=c+d$, which of the following is equivalent to the expression $x^2-c^2-2cd-d^2$? A. $(x+a)^2$ B. $(x-a)^2$ C. $(x+a)(x-a)$ D. x^2-ax-a^2	C
3.17	e117d3b8	If *a* and *c* are positive numbers, which of the following is equivalent to $\sqrt{(a+c)^3}\cdot\sqrt{a+c}$? A. $a+c$ B. a^2+c^2 C. $a^2+2ac+c^2$ D. a^2c^2	C
3.18	42f8e4b4	One of the factors of $2x^3+42x^2+208x$ is $x+b$, where b is a positive constant. What is the smallest possible value of b?	8

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3.19	433184f1	Which expression is equivalent to $\frac{4}{4x-5} - \frac{1}{x+1}$? A. $\frac{1}{(x+1)(4x-5)}$ B. $\frac{3}{3x-6}$ C. $-\frac{1}{(x+1)(4x-5)}$ D. $\frac{9}{(x+1)(4x-5)}$	D
3.20	c3b116d7	Which of the following expressions is(are) a factor of $3x^2 + 20x - 63$? I. $x - 9$ II. $3x - 7$ A. I only B. II only C. I and II D. Neither I nor II	B